
CUTTLESW!SH
HELPING ELDERLY COOKS

Initial Redesign

Selected Concept Rationale

Our original design was a continuation of the StirMaster Pro, a design which came to fruition from our initial ideation, inspired by our previously selected brand, KitchenAid.

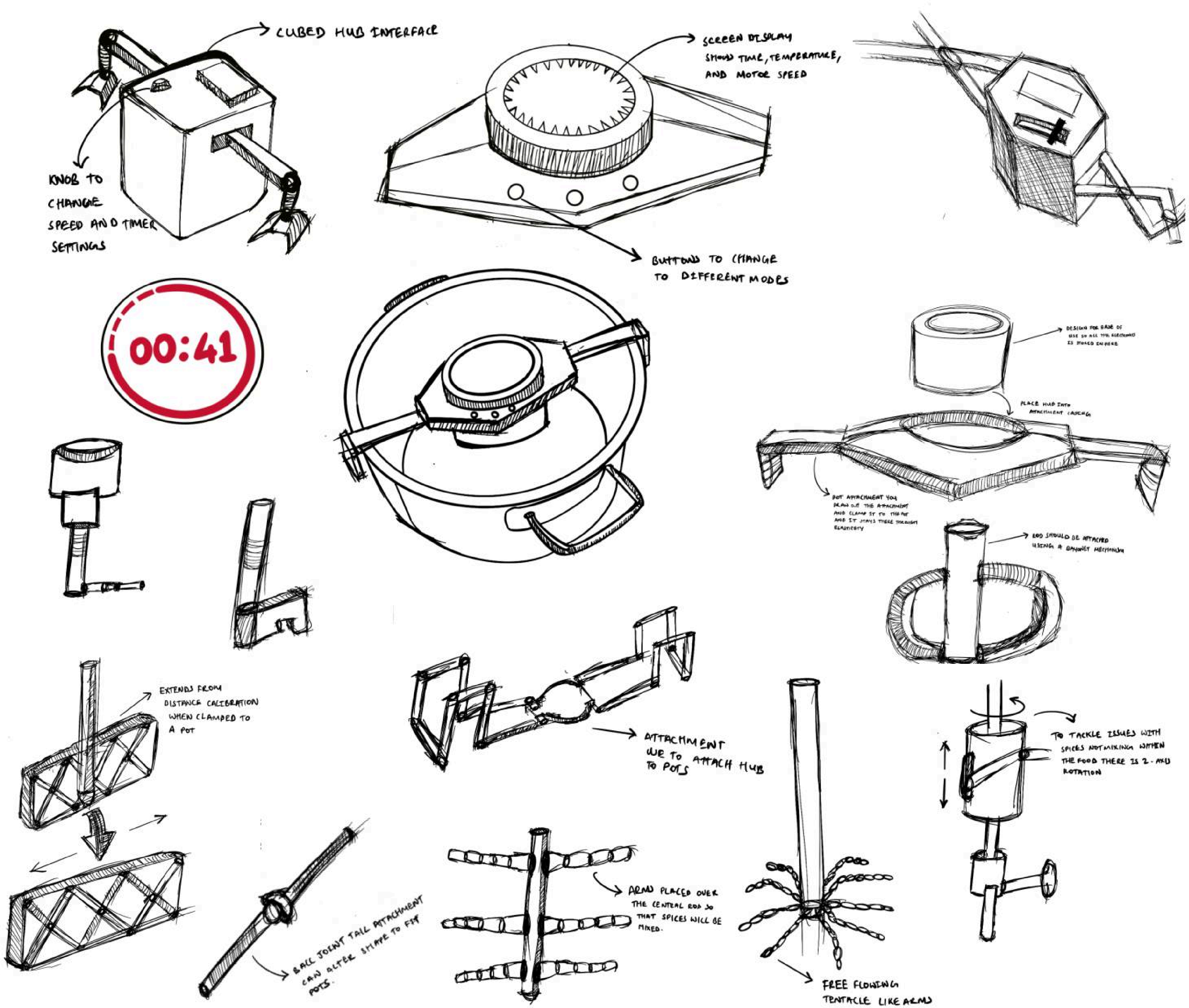
Issues that arose early on were to do with the form of the product with the placement of the screen as buttons over the centre of a hot pot may be uncomfortable for the user during setup. Moreover the screen may prove both difficult to view and experience malfunctions with the combination of steam and heat.

Onto the interfaces and solutions:

Initial designs were focused on improving mechanical and UI features to see in which way we wanted to lead the design, introducing things like the circular hub in the middle to hold electronics, adding additional arms to support the design more, and introducing features like a knob as input.

Having taken inspiration from a watch face, the interface was adapted to operated on a rotary encoded gear such that the outer ring of the central hub dependent on which mode was pressed would be able to regulate speed and duration of stirring.

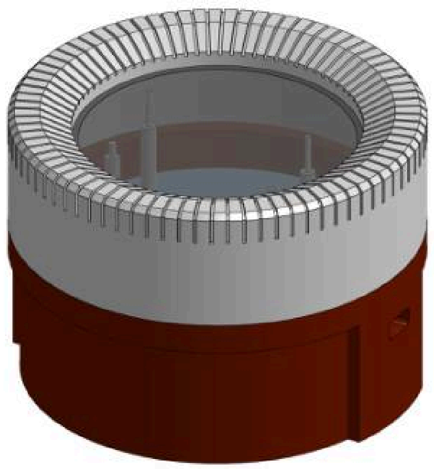
This paired visual feedback of an LED ring provided a much more elegant and effective interface for our target market to operate.



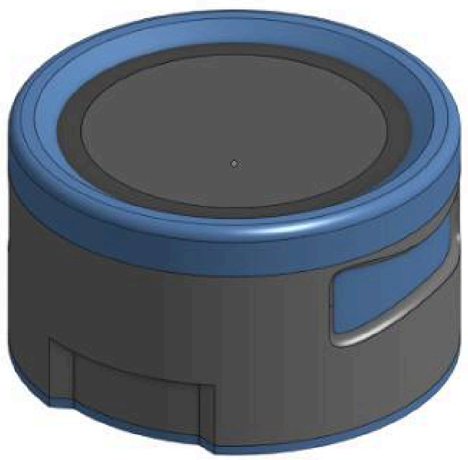
CUTTLESW!SH

Automatic Stirrer

The final design was arrived to through a long development process and concluded in the cute, zoomorphic appearance that was made to reflect the look of a cuttlefish. The design came about from version 4, where we experimented with having “tentacle-like” attachments and arms to give more flexibility and customisability to the height and shape of the pot and arm it could fit in. This was the developed from physical joints, to bendable silicone and flexible support arms to reduce the required effort from cleaning. UI design also shifted throughout from touchscreens, to segmented displays and finally LED rings and buttons.



Version 1
KitchenAid



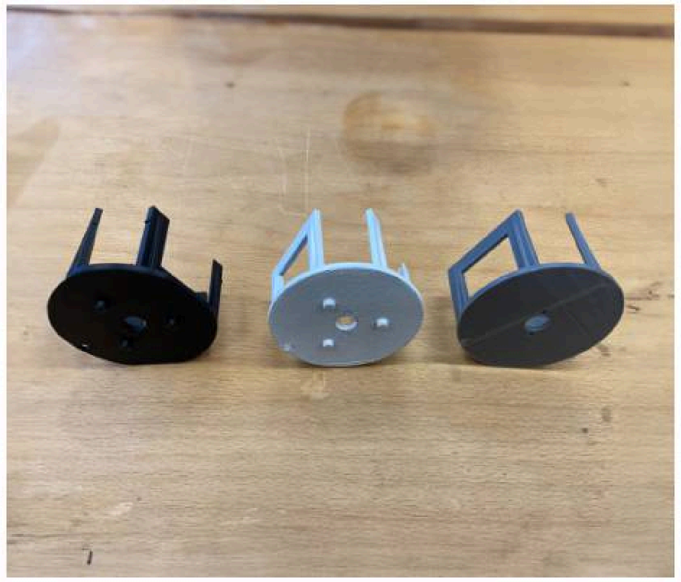
Version 2
JosephJoseph

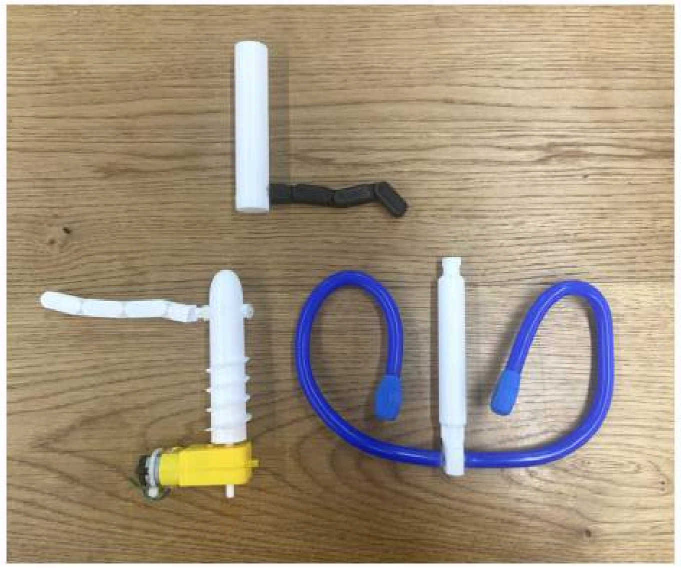
Version 3
JosephJoseph

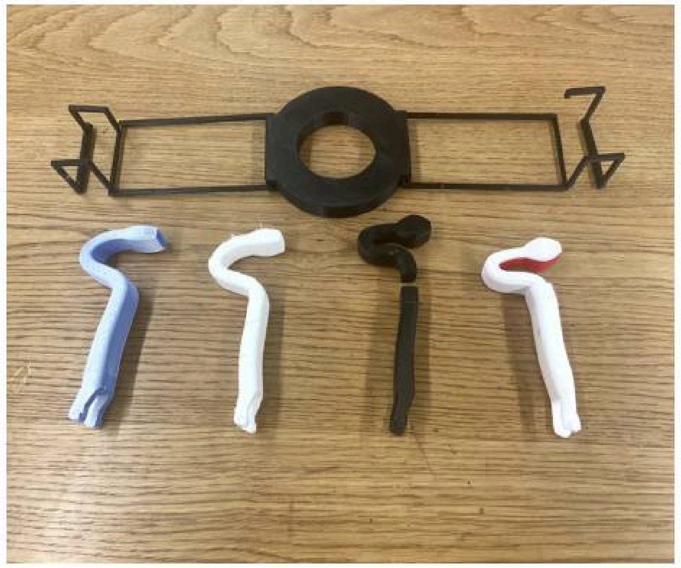
Version 4
KitchenAid

Version 5
PELEGDES!GN

CONCEPT DEVELOPMENT







Issues & Solutions

Mechanical Issues

The main mechanical issues came about when focusing on the support mechanism. Firstly, there were issues with the legs bending unevenly, which were fixed by introducing a detent mechanism in the joint.

Then, there were issues with ensuring the mechanism would attach to the pot and grip it. Testing with TPU models allowed us to find that although it would grip well enough, having the hole arm made of TPU would make it bend too much, leading to us making just the end out of TPU.

Electronic Issues

Electronics plagued us throughout this project. Between taking long to assemble and build, and then soldered circuits continuously failing, testing and development took very long. The final model ended up having to be assembled using pin wires and could not be soldered as originally planned.

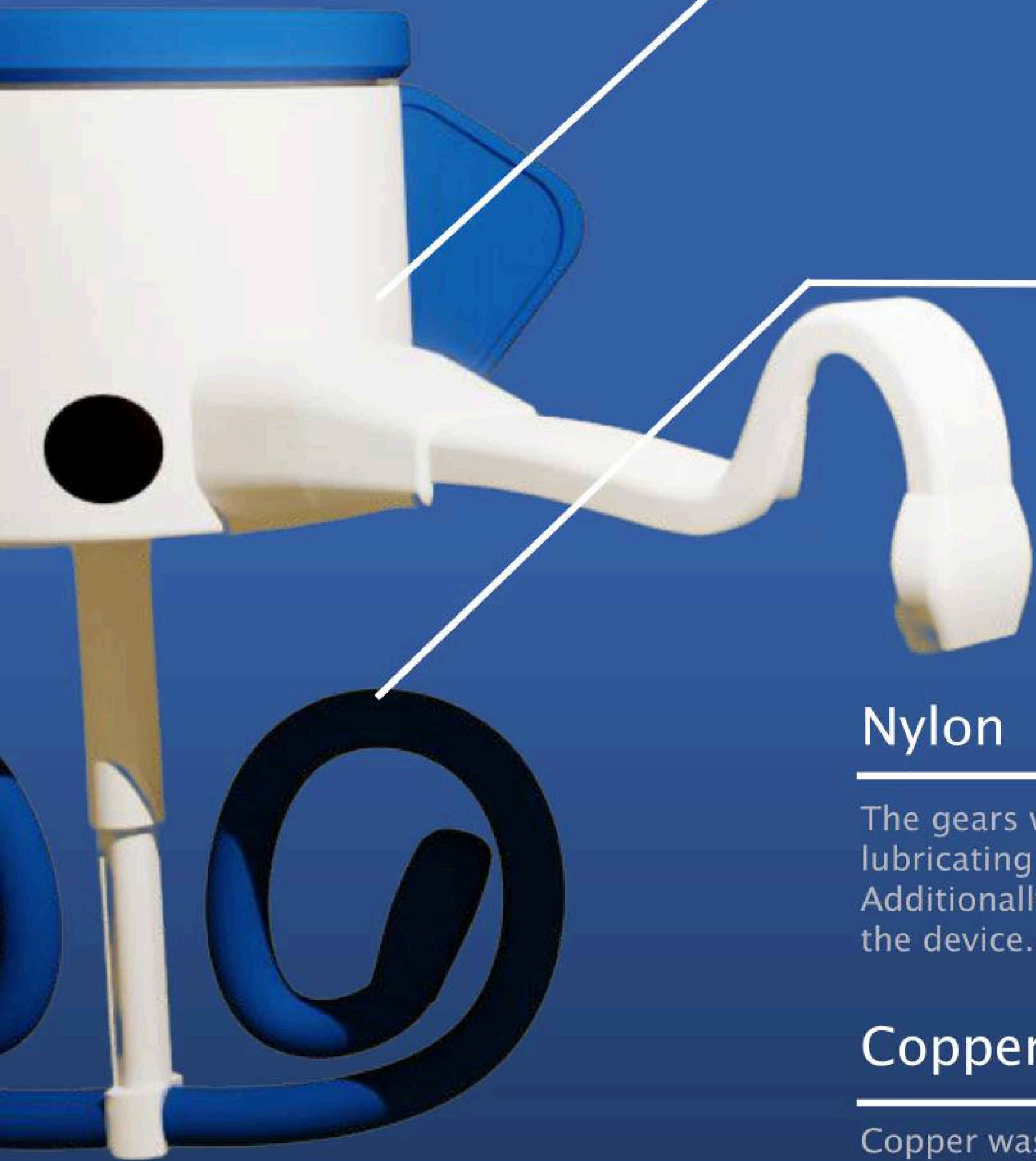
Heat, water and food safety

We had to keep in mind throughout the design process how we would protect the electronics from the heat, steam and even splashes of food, whilst making sure it was still food same. This was done through smart material choice and having multiple layers to protect the electronics as much as possible from external factors.

COLOURS, MATERIALS & FIN!SHES

SEMIOTIC ANALYSIS

The Cuttlesw!sh embodies a thoughtful selection of colours, materials, and finishes, ensuring not only functionality but also aesthetic appeal. The chosen materials contribute to the product's longevity and performance, while the smooth finishes add a touch of sophistication, making the Cuttlesw!sh a delightful addition to any kitchen. With the main body being kept white, the blue highlights can be changed to whatever colour the user desires.

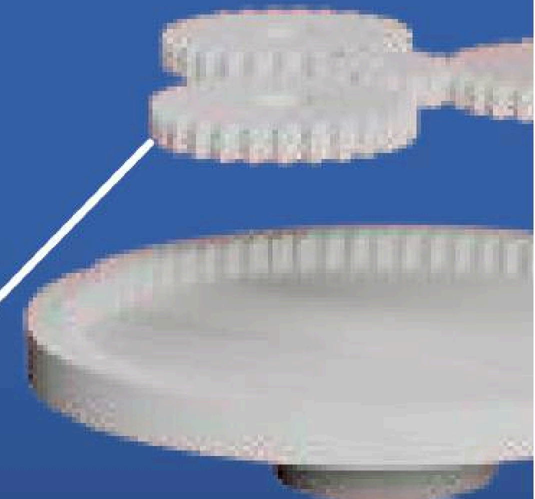


ABS

This is a versatile material which is both temperature resistant and, more importantly, food-safe. Its durable nature and ability to be injection moulded further adds to our desire to use it.

Silicone

Silicone has been used in multiple aspects of the stirrer for its mouldable and food-safe properties. It covers the stirrer, ensuring hygiene and safety, seals the support arm joints to prevent food and dirt ingress, provides a grip on the support arms for better adherence to the pot, and protects the electronics on the top by sealing them securely.

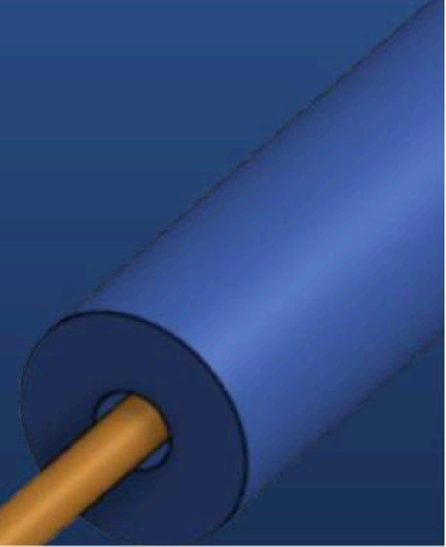


Nylon

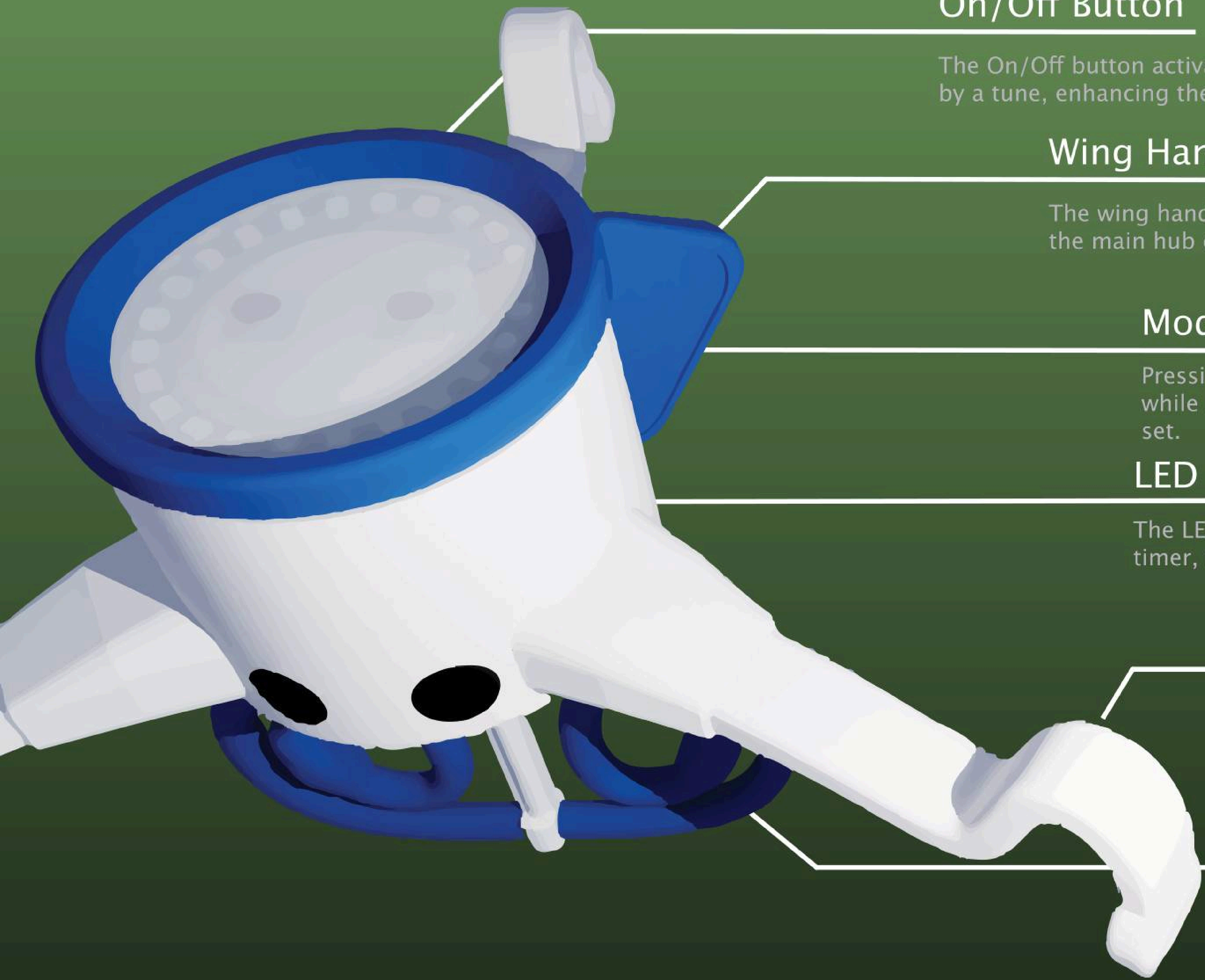
The gears were designed to be made of nylon due to its self-lubricating properties, eliminating the need for additional grease. Additionally, nylon's durability ensures long-lasting performance in the device.

Copper

Copper was incorporated inside the mouldable stirrer because it provides the necessary flexibility to bend while retaining its shape during cooking.



The Cuttlesw!sh™ has been designed to ensure a simple yet highly functional user experience. Every element, from the intuitive controls to the adaptable components, is crafted to enhance usability. These features detailed below illustrate how the device meets the needs of users to make cooking accessible.



On/Off Button

The On/Off button activates the device and plays a brief LED animation accompanied by a tune, enhancing the user experience with visual and auditory feedback.

Wing Handles

The wing handles are ergonomically designed to allow users to easily lift the main hub off the pot, providing a secure and comfortable grip.

Mode & Start Button

Pressing this button cycles through the three available modes, while holding it starts the Cuttlesw!sh only when a timer has been set.

LED Ring

The LED ring serves as the primary output display, indicating the timer, temperature, and motor speed through different colours.

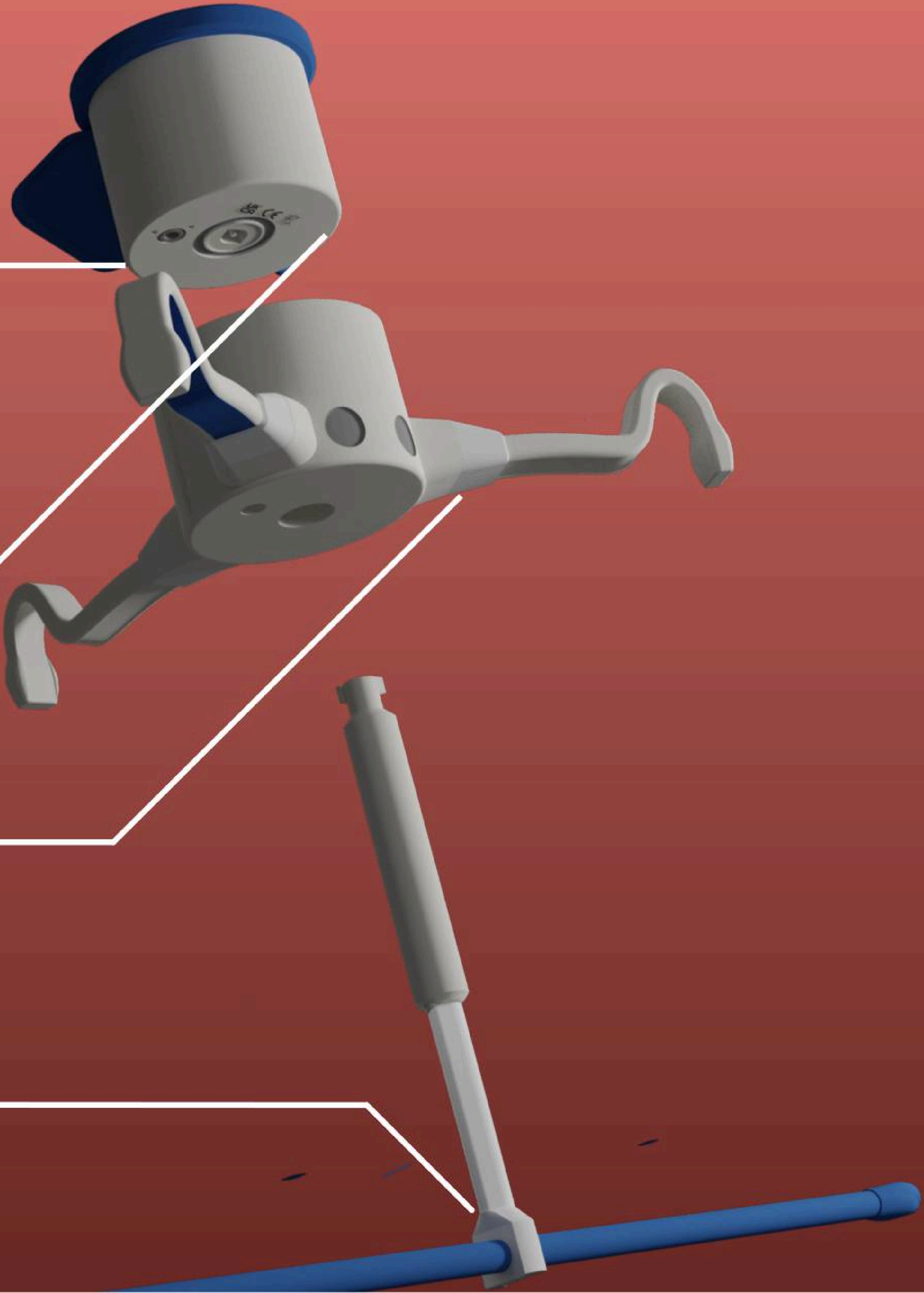
Flexible Arms

These flexible arms can clamp onto various surfaces, ensuring a secure attachment and stability during operation, adapting to different pot shapes and sizes.

Mouldable Stirrer

The mouldable stirrer allows users to customise the stirring attachment to fit the specific size and function required.

The assembly of the Cuttleswish™ is designed to be straightforward and efficient, with key components that ensure robust functionality and ease of use. Below, we outline the critical features and methods used in the product's assembly.



3 Key Sections

The product is made up of 3 key components, the main hub which houses all the electronics, the main shell which protects the hub and holds it above the pot, and the attachment to stir the food.

Snap & Press Fittings

The majority of the components have been designed to be assembled by snap and press fittings allowing for us to not use more permanent attachment methods.

Screws

Some components were made to be screwed on, such as the motors and start of arms.

Provisioning

Three key ideas allow the different components to fit together: Clearance fits, Interference fits and Allowance & Enveloping.

BRANDING CONSIDERATIONS

Kitchenaid

Known for its stylish and sleek finish, Kitchenaid produces high-quality kitchen appliances that are both functional and visually appealing. [1]

Artisan Mixer Tilt-head 4.8L

Core Values

Quality
Emphasis on producing durable, high-quality products.

Innovation
Continual improvement and innovation in kitchen solutions.

Community
Commitment to supporting and building community connections.

Heritage
Creating a strong legacy of crafts-manship and design.

Est 1919

Market Value of $12.4 billion, 2024 [2]

JosephJoseph

Specialising in easy-to-use, minimalist kitchen tools with a splash of colour, catering to practical and visually engaging needs. [3]

Helix Garlic Press

Core Values

Intentiveness
Creating products that solve everyday problems in original ways.

Determination
Persistence in bringing daring and distinctive products to market.

Independence
Following their instincts to innovate in the homeware industry.

Collaboration
Emphasis on teamwork to ensure functionality and aesthetic appeal.

Est 2003

Market Value of $54 million, 2024 [4]

Ototo

Renowned for its creative, quirky kitchen gadgets that blend functionality with a playful, light-hearted aesthetic. [5]

Shelly Measuring Cups

Core Values

Playfulness
Incorporating fun and humor into everyday kitchen items.

Functionality
Ensuring that products are not just whimsical but also practical.

Innovation
Continuously exploring new ideas and pushing design boundaries.

Accessibility
Creating user-friendly products that are accessible to all customers.

Est 2004

Market Value of N/A, 2024

BRANDING IMPLEMENTAT!ON

PelegDes!gn

Shahar Peleg created Peleg Design in 2004 and it is well known for its playful and practical kitchen tools. They make tedious tasks enjoyable with whimsical designs and clever product names blending putting the fun in functionality.

Wine Bottle Holder

Soap Opera

Wine Bottle Holder

Soap Opera

Peleg Design stands apart from companies like Kitchenaid, Joseph Joseph, and Ototo because of its simple, elegant, yet witty design language. Peleg's minimalistic concepts, in contrast to Ototo's: humorous yet less refined. This playful aesthetic broadens our target market for Cuttleswish to be a product for both the elderly and younger generation. [6]

Brand Iceberg

Humour

Shahar Peleg infuses wit into all products from their design down to the name.

Functionality

"Our products are not just decorative; they are meant to be used every day."

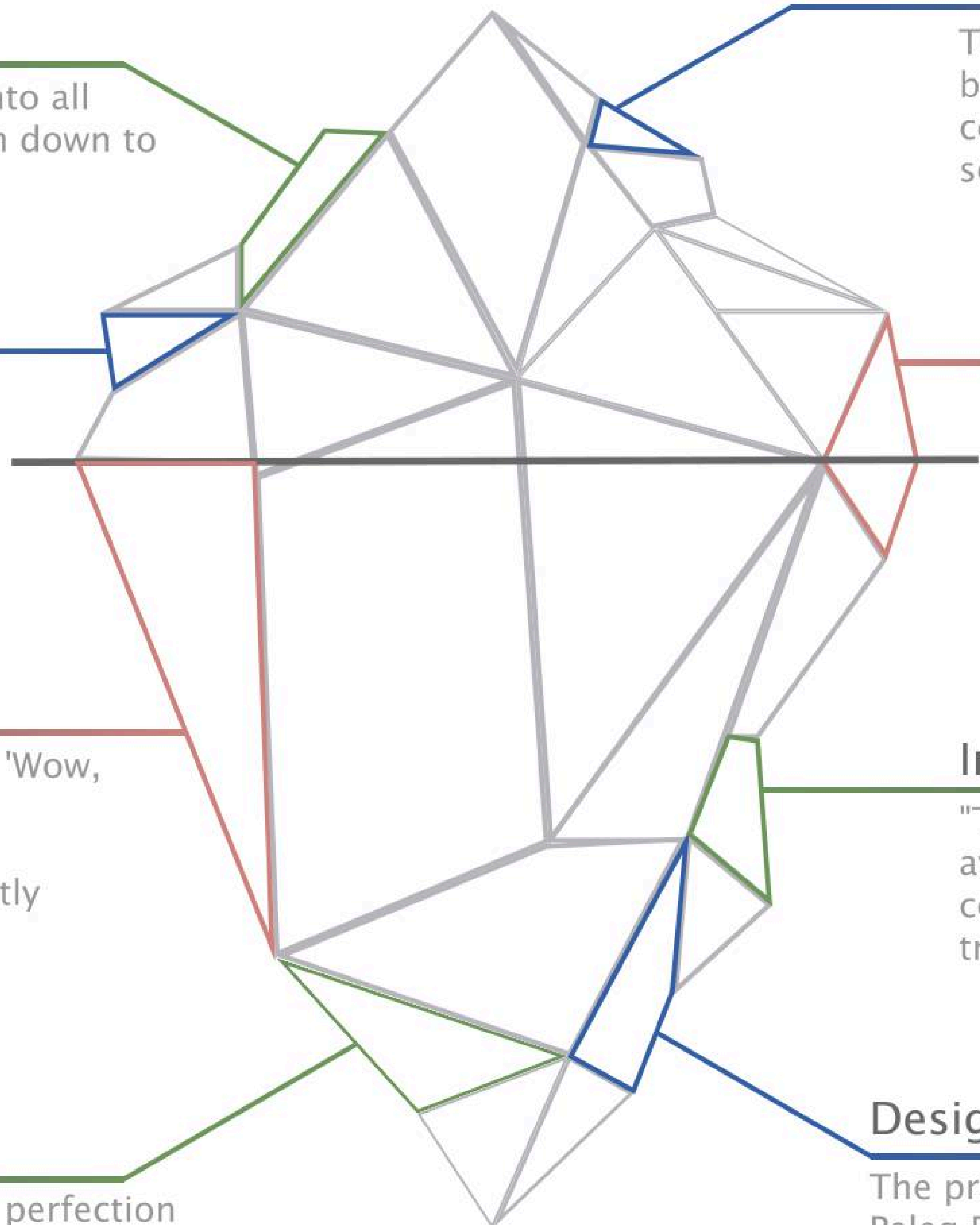
User Engagement

"We love it when people say, 'Wow, this is brilliant!'"

They make products to directly align with consumer needs.

Attention to Detail

Every product is design to perfection from physical product to branding



Visual Identity

Their formal design language can be seen in their typeface but is contrasted by their playful logo as seen with the inverted "i".

Aesthetics

"Known for their whimsical designs and pun-based names... (Shahar) invites the viewer to take a closer look, revealing a fresh view of everyday products."

Innovation

"Transforming what was once an average, been-there-done-that concept into an innovative and truly exciting experience."

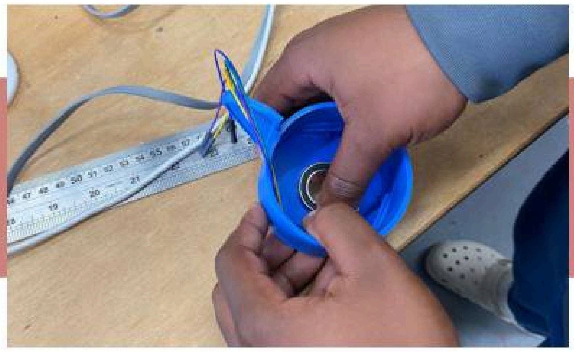
Design Inspiration

The primary source of inspiration for Peleg Design's designs is zoomorphism, which gives their creations vitality.

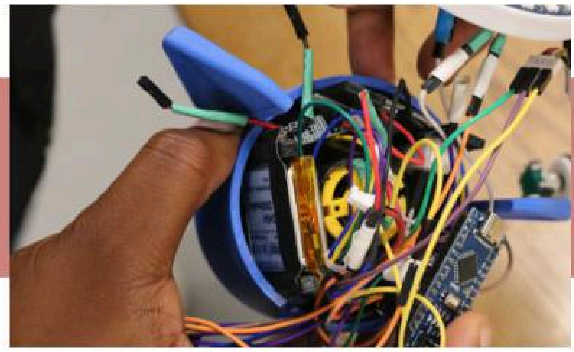
PRODUCT ASSEMBLY PROCESS

TIMINGS & COST ESTIMATE

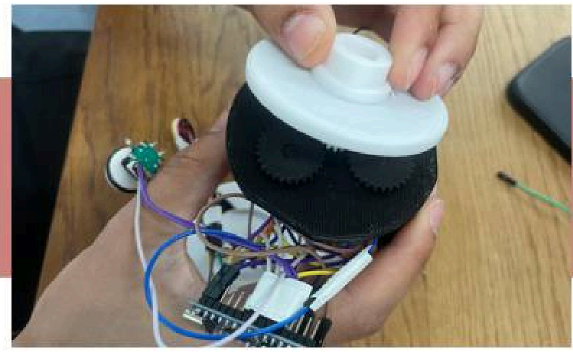
01 Inner Shell



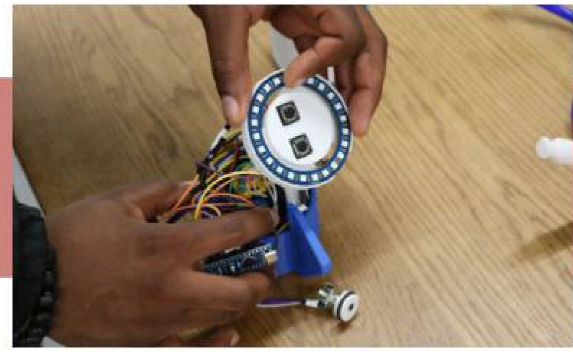
20sec – Press fitting the bearing and IR sensor in place.



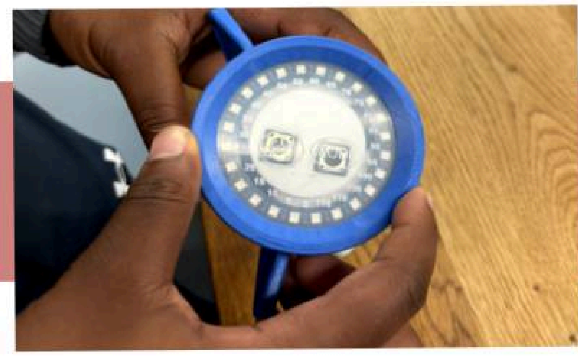
15mins – Attaching the electronics to the main base.



1min – Inserting the gears.

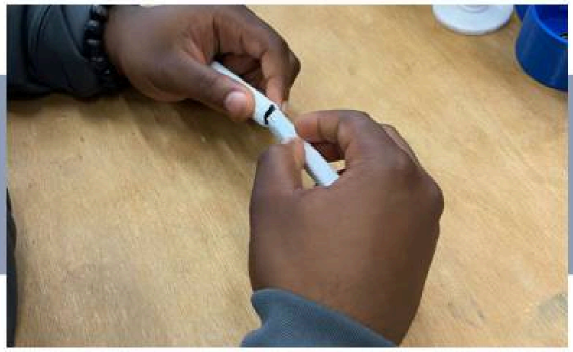


5mins – Assembling everything in the main shell.

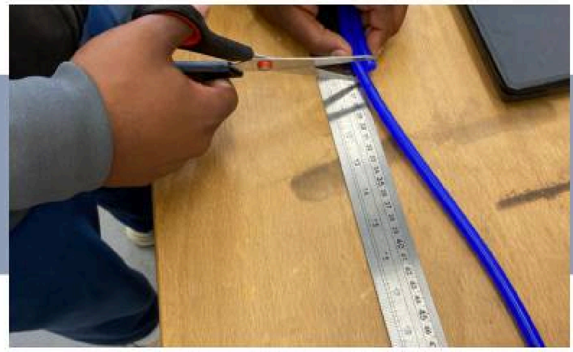


2mins – Attaching the silicone cover and top knob.

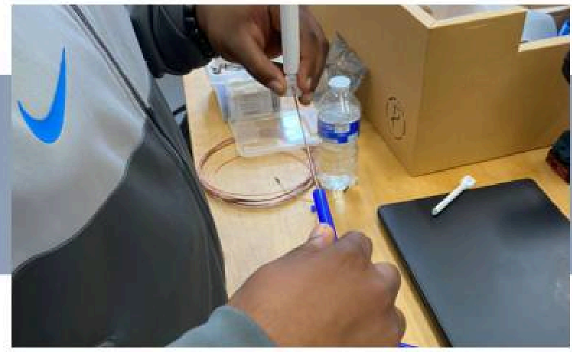
02 Attachment



20 sec – Snap fitting the bottom rod to the top.

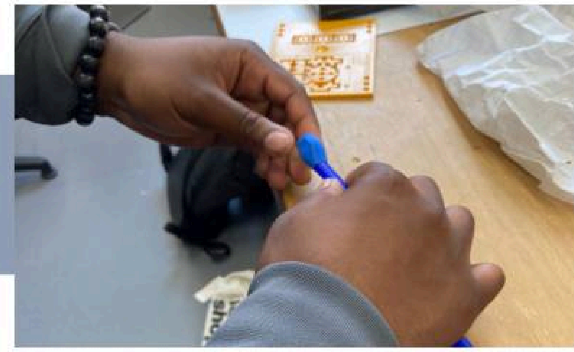


3 mins – Measuring and cutting the silicone & copper.



20 sec – Inserting the copper and silicone.

5 mins – Glueing everything to the bottom attachment.



5 mins – Glueing the end of the tube stopper.

03 Outer Shell

1 min – Snap fitting the main arm to the outer shell.

5 mins – Glueing the TPU ends to the support arms.

10 sec – Inserting the inner shell to the main one.

20 sec – Putting the attachment in.

5 sec – Place it on the desired pot.

Total Time: 43 minutes 40 seconds

The actual total assembly time will be significantly reduced in a manufacturing setting due to the use of automated processes and machinery.

SYSTEM ARCH!TECTURE

POWER CALCULATIONS & COMPONENTS

Calculations

This diagram details the Cuttleswish components and power calculations outlining requirements for suitable performance.

01 Arduino Nano

Power Required: 1 W
Current Rating: 200 mA

02 Infrared Sensor

Power Required: 0.075 W
Current Rating: 15 mA

03 NeoPixel Ring

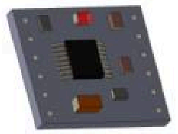
Power Required:2.16 W
Current Rating: 432 mA

04 Speaker

Power Required: 1 W
Current Rating: 200 mA

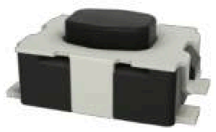
05 DC Motor

Power Required: 4 W
Current Rating: 700 mA



06 H-Bridge

Power Required: 0.5 W
Current Rating: 100 mA



07 Buttons

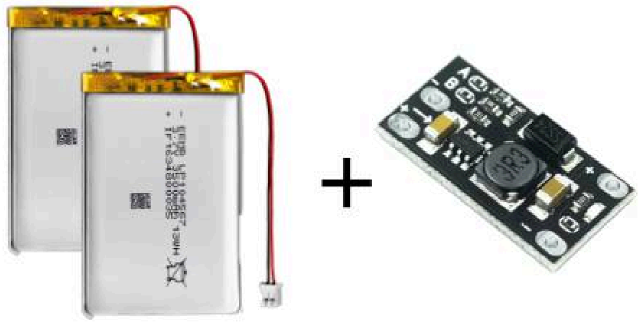
Power Required: 0.01 W
Current Rating: 10 mA

08 Encoder

Power Required: 0.25 W
Current Rating: 50 mA

Total Current Draw: 1707mA

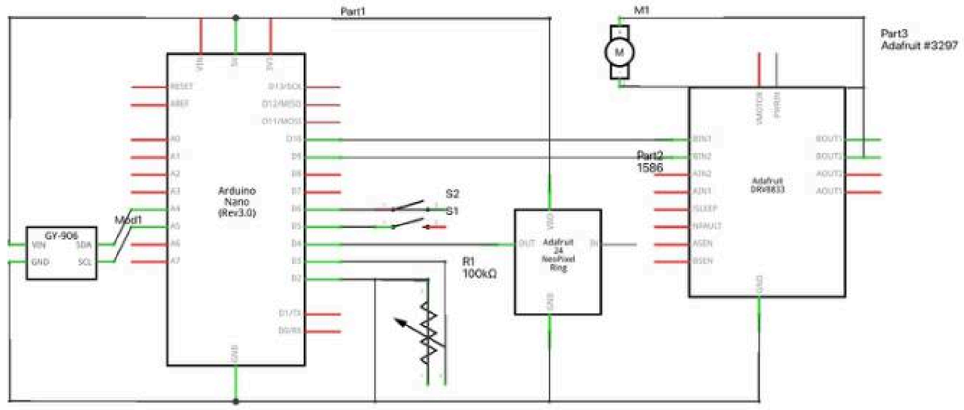
Battery Capacity=1.71×4=6.8Ah



To ensure the CuttleSw!sh can operates for 4 hours, the total current draw was multiplied by 4. Due to difficulty finding a single compact battery, two 3.7V, 3.4 Ah lithium batteries with a step-up were sought to achieve the 5 V and 6.8Ah requirement

Architecture

The schematic and pseudo code illustrate the detailed circuitry and logic controlling the Cuttleswish.



System Schematic

```
SETUP {
    init components, NeoPixel, MLX90614;
}

LOOP {
    checkButton();
    IF power button pressed { togglePower(); }

    IF alarm {
        IF alarm duration > 5s { shutdown(); }
        ELSE { playAlarm(); }
        RETURN;
    }

    IF timer { updateTimer(); }

    IF active AND debounce delay {
        IF temperature mode { updateTempPixels(); }
        ELSE { handleEncoder(); }
        lastUpdate = millis();
    }
}

// Functions
FUNCTION togglePower {
    IF active { shutdown(); }
    ELSE { turnOn(); colorIndex = 0; playMelody(powerOnMelody); updatePixels(); }
}

FUNCTION handleEncoder {
    tmpdata = readEncoder();
    IF tmpdata {
        IF motor speed mode { blueCounter += tmpdata; updateMotor(); }
        IF timer mode { greenCounter += tmpdata; updateTimerPixels(); }
    }
}

FUNCTION readEncoder { RETURN encoder change; }

FUNCTION updatePixels {
    IF temperature mode { updateTempPixels(); }
    ELSE IF motor speed mode { updateMotorPixels(); }
    ELSE IF timer mode { updateTimerPixels(); }
}

FUNCTION updateTempPixels { temp = readTemp(); setPixels(red, temp); }

FUNCTION updateMotorPixels { setPixels(blue, blueCounter); }

FUNCTION updateTimerPixels { setPixels(green, greenCounter); }

FUNCTION updateTimer {
    elapsed = millis() - timerStart;
    remaining = greenCounter - elapsed / 60000;
    IF remaining <= 0 { shutdown(); playAlarm(); }
    IF timer mode { updateTimerPixels(); }
}

FUNCTION updateMotor {
    IF timer { pwm = map(blueCounter); analogWrite(MOTOR_PIN_IN1, pwm); digitalWrite(MOTOR_PIN_IN2, LOW); }
    ELSE { stopMotor(); }
}

FUNCTION playMelody(melody) {
    FOR each note { tone(note); delay(); noTone(); }
}

FUNCTION playAlarm {
    FOR each note in alarmMelody { tone(note); delay(); }
}

FUNCTION clearPixels { FOR each pixel { pixels.setPixelColor(off); } pixels.show(); }

FUNCTION shutdown {
    active = false; timer = false; alarm = false;
    noTone(12); playMelody(shutdownMelody); clearPixels(); stopMotor();
}

FUNCTION checkButton {
    IF button1 pressed { colorIndex++; IF colorIndex > 2 { colorIndex = 0; } playMelody(beepMelody); updatePixels(); }
    IF button2 long press AND timer mode AND greenCounter > 0 AND !timer {
        timer = true; timerStart = millis(); playMelody(timerStartMelody); updateMotor(blueCounter);
    }
```

Pseudo Code

PRODUCT COMPL!ANCE

UK & EU

Peleg Design products in the UK conform to both UK and EU requirements.

The UKCA marking certifies that our product is in compliance with UK safety regulations, meeting health, safety, and environmental standards.

The CE marking ensures adherence to EU safety, health, and environmental regulations allowing our product to be freely traded in the EU

Declaration

As part of ensuring the highest quality and safety, the Cuttleswish™ must include a declaration of conformity.

These document detail the manufacturer's name, address, and relevant product information, confirming that the product meets all applicable UKCA and CE standards. This documentation can be found in the user guide.

Implementation

Known for its stylish and sleek finish, KitchenAid produces high-quality kitchen appliances that are both functional and visually appealing.

RoHS
(Restriction of Hazardous Substances)
Ensures safe, eco-friendly materials, limiting harmful substances.

EMC/EMF
(Electromagnetic Compatibility and Fields)
Meets standards for effective, interference-free operation.

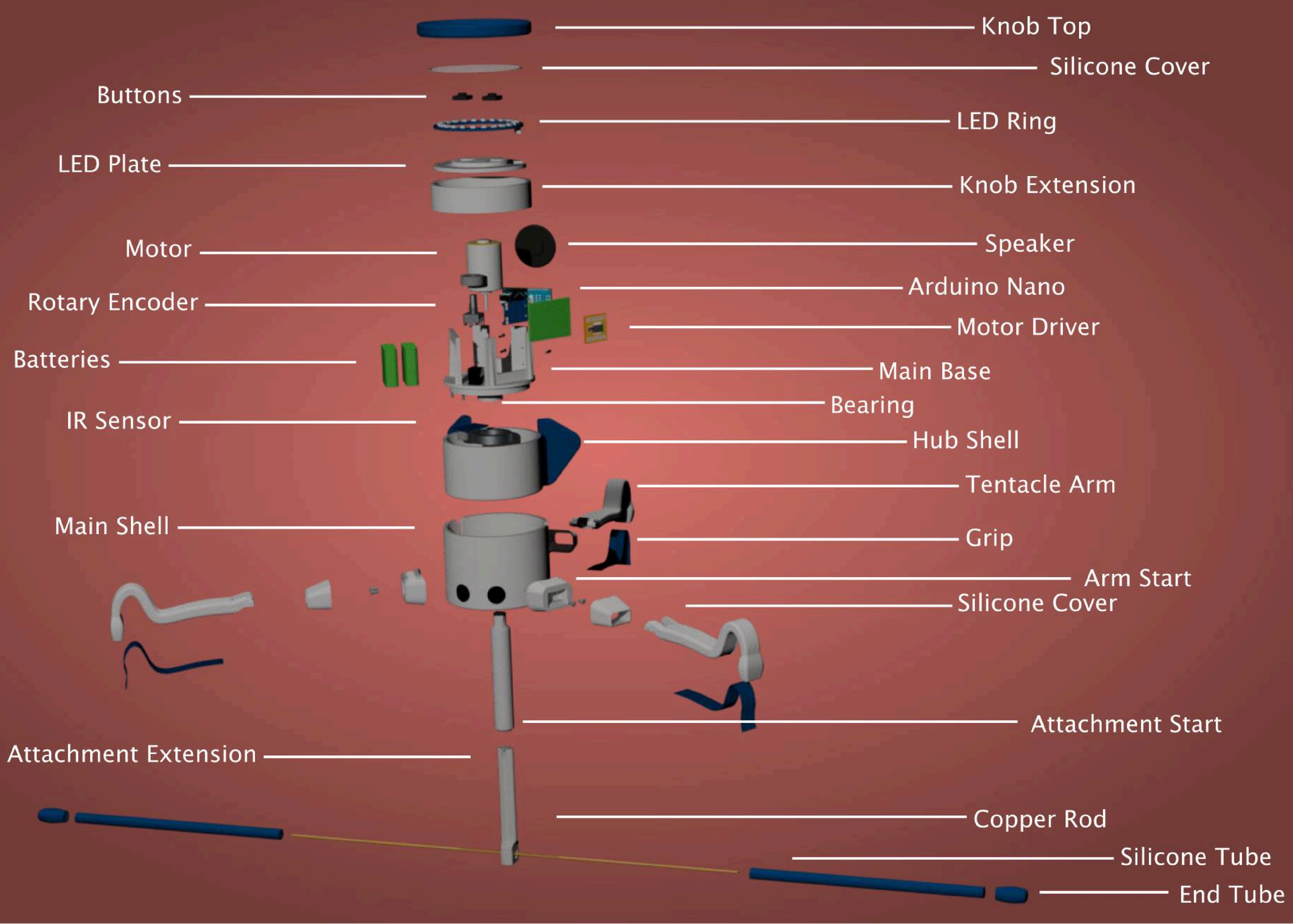
WEEE
(Waste from Electrical and Electronic Equipment)
Ensures safe disposal and recycling of electronics.

ERP
(Energy Related Products)
Guarantees energy efficiency and low environmental impact.

FCM
(Food Contact Materials)
Ensures food-contact materials release no harmful substances.

EXPLODED V!EW

VISUAL RENDER



PRODUCT LABELL!NG

NAME & MARKINGS

CUTTLESW!SH

Taking inspiration from Peleg Design’s witty and humorous product names, we brainstormed Cuttlesw!sh, a cuttlefish which ‘swishes’ your food.

UKCA & CE

The UKCA (UK Conformity Assessed) and CE marks are mandatory labels that must be affixed to the product. These marks indicate compliance with relevant UK and EU regulations, ensuring the product meets safety, health, and environmental standards. [7]

WEEE

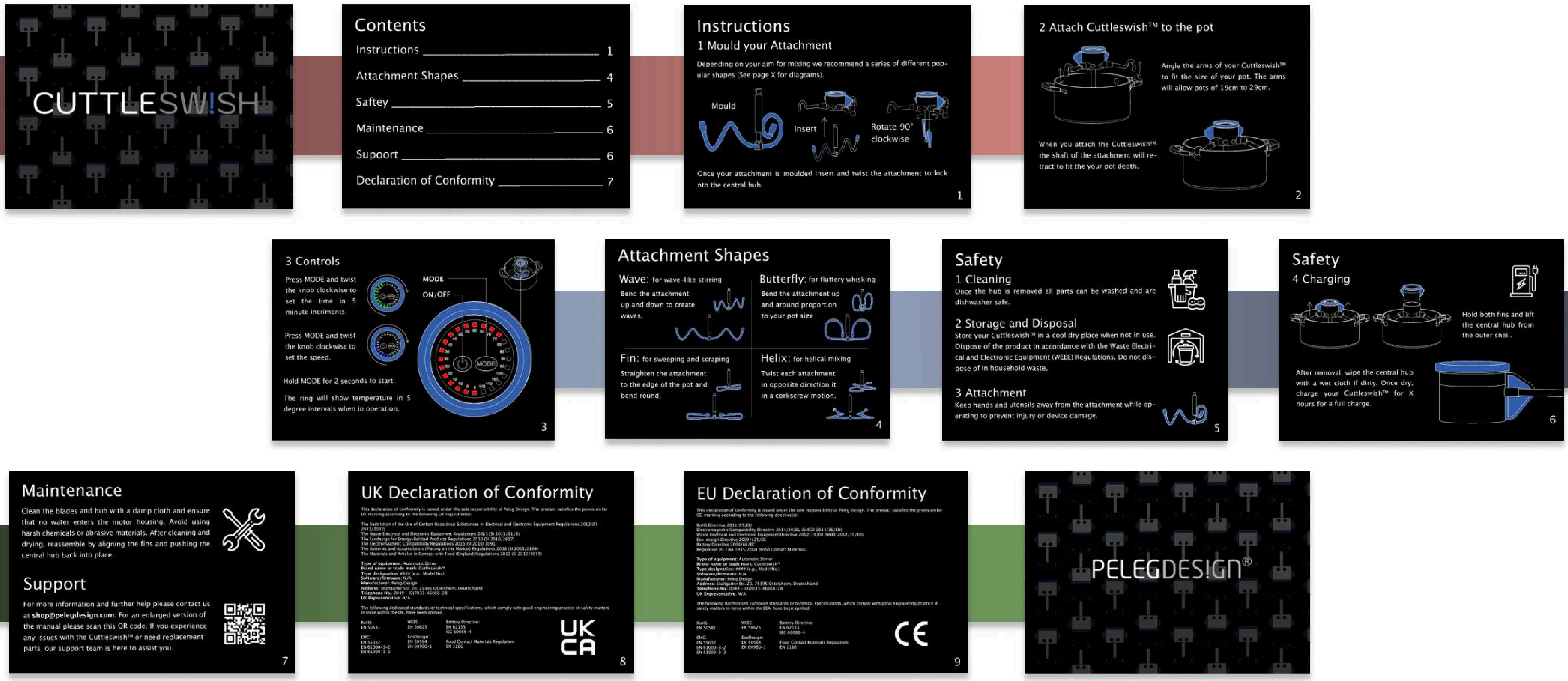
The Waste Electrical and Electronic Equipment (WEEE) directive focuses on reducing the environmental impact of electronic waste. Users are encouraged to recycle their Cuttleswish responsibly at end of life, opposed to disposing in household waste.

Food Contact Materials

The food-safe symbol signifies that the materials used in manufacture are safe for use with food, adhering to regulations that limit harmful substances.

USER GUIDE DES!GN

INSTRUCTION MANUAL



Instructions p. 1-3

Detailed step-by-step guide for assembling and using Cuttleswish with descriptions and infographics, ensuring user-friendly operation and ease of use.

Attachment Shapes p. 4

Overview of various mouldable silicone attachments, highlighting their adaptability to different pot shapes and sizes for efficient mixing.

Safety p. 5-6

Comprehensive safety guidelines and warnings to ensure safe operation and prevent accidents, crucial for user protection.

Maintenance and Support p. 7

Tips for maintaining Cuttleswish and support information, ensuring longevity and provide customer assistance, including a QR code for a digital enlarged guide.

PACKAGING DESIGN

EXPLODED VIEW

The Cuttleswish packaging design facilitates an engaging unboxing experience, highlighting the product's quality and thoughtful details. Each layer, from the outer box to the base compartment, is designed for ease of use, protection, and visual appeal. This design not only protects the product but also delights the user from the first moment they open the box.

Outer Box

Made of high-quality non-corrugated cardboard. This features branded graphics and instructions, created using die-cutting and printing processes.

Labelling

Packaging is clearly labeled with the product name, description, company logo, and regulatory symbols: UKCA, CE, WEEE and FCM. (Package Disposal symbol are featured on the underside.)

The QR code featured provides easy access to the user manual and support information.

Protective Foam Layer

Constructed from HDPE foam, this insert secures the Cuttleswish during transport. It is precisely moulded and cut for a snug fit, ensuring protection from damage.

Base Compartment

The base compartment is made from durable cardboard and securely stores the USB charging cable, user guide and mouldable attachment.

It features moulded slots for organisation and protection, ensuring stability and ease of access during transit. Manufactured using precise die-cutting, it combines functionality with sturdiness.

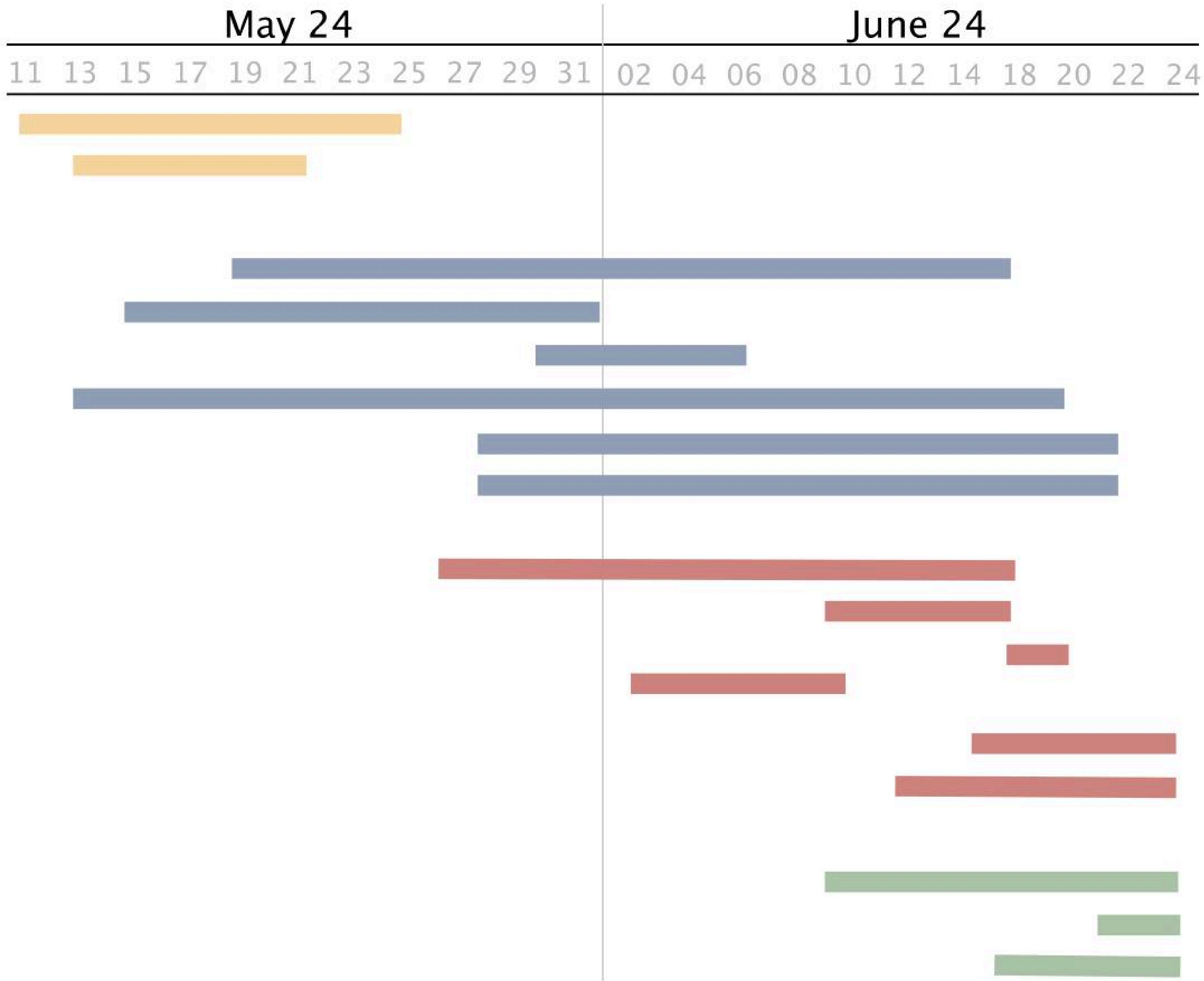
Branding Wrap

Made from printed tissue paper, designed with a tessellation of Cuttleswish icons . It is produced through printing and die-cutting to fit neatly around the packaging.

Cuttleswish™
Automatic Stirrer for Elderly
PELEGDESIGN®
CUTTLESWISH
Step 1
Step 2
Attach Cuttleswish™ to your pot
The three arms snap to fit pot diameters 19cm and 29cm to cater to your biggest and smallest pots.
The shaft height will adjust when attaching your Cuttleswish™ to ensure the attachment is the perfect height for your pot.
Step 3
Set time, speed and go!
Twist the knob when on green to set the time.
Each LED represents 5 minutes of stirring.
Twist the knob when on blue to set the speed.
Then sit back, relax and let Cuttleswish™ do the heavy lifting.

Gantt Chart

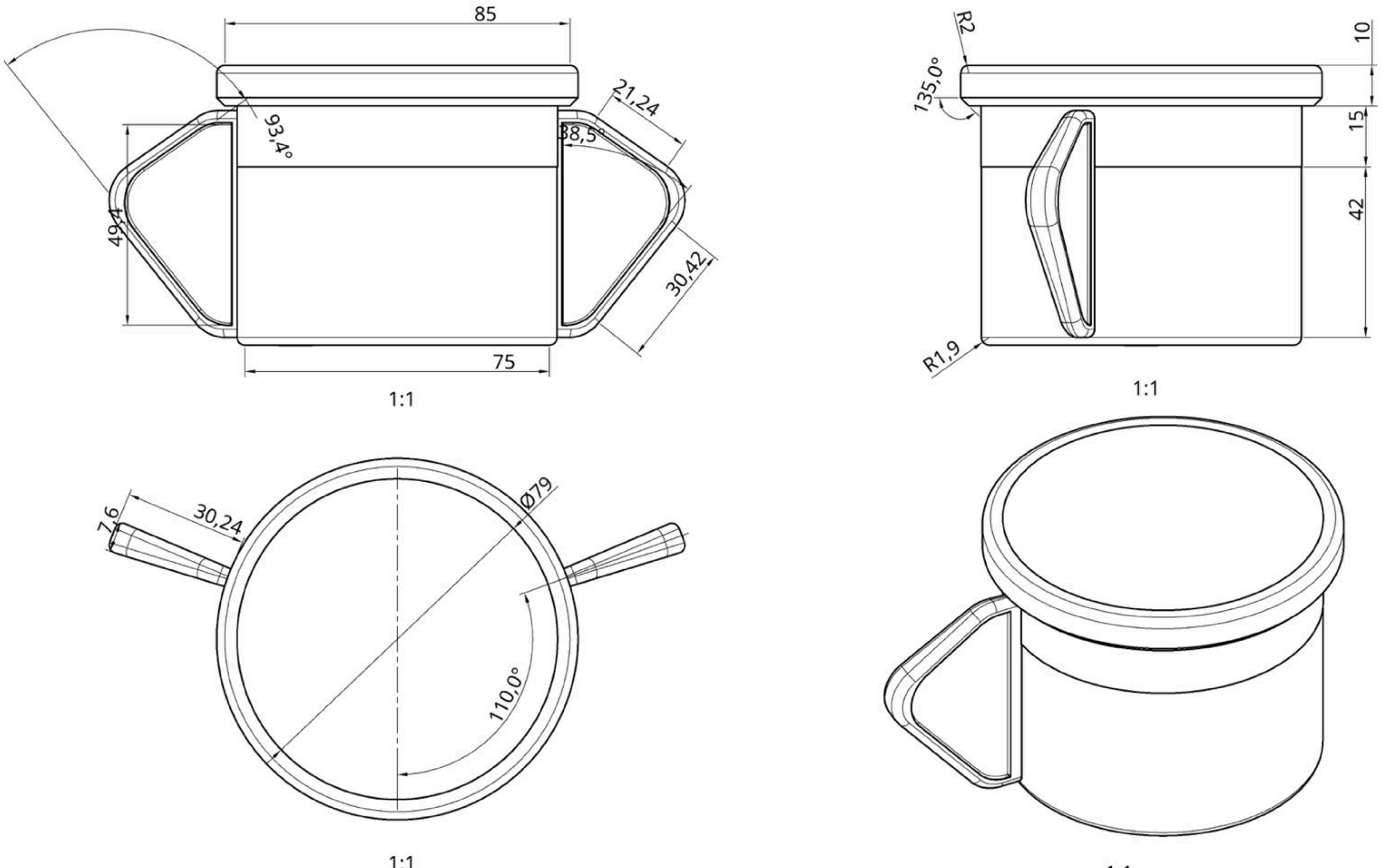
Task	Completed by
Concept Ideation	
Initial Sketching	A, F, S, E
Concept Selection	A, F, S, E
Development	
Iterations & Improvement	S, F, E
Research of Brands	S, F, E
Implementation of Brand Influence	F, S
Digital Prototyping	F, S
Physical Prototyping	S, F, E
Electronics Prototyping	E
Finalise	
Final CAD Model	F
DFMA	F
Technical Drawings	F
Components Purchasing	S
Production of Final Design	S, F, E
Final Renders	F, S, E
Delivery	
Final Report	S, F, E
Final Presentation	S, F, E
Video Production	F



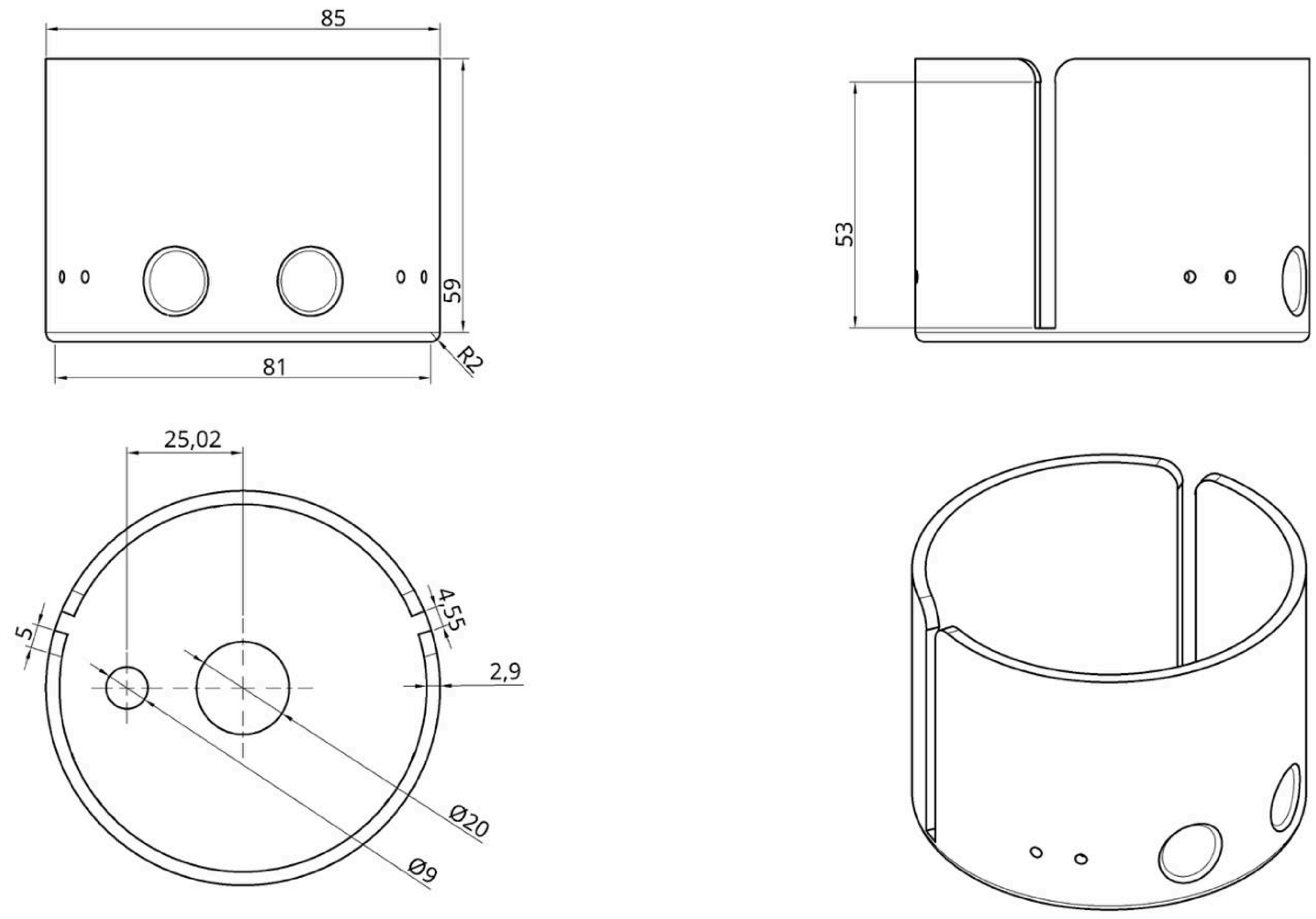
Weekly meetings proved vital in the early stages for our divergent stages of concept development. These meetings ensure that we consistently on track with weekly tasks.

Conducting initial ideation together followed ensured we all had equal input into the final concept. From this point onwards delegating tasks proved the most efficient way for steady progress.

Despite facing multiple challenges and obstacles with team dynamics. With perseverance and teamwork we were able to complete deliverables to the best of our ability.



UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN MILLIMETERS ANGULAR = ± °		NAME	SIGNATURE	DATE	
	DRAWN	FRANCESCO RASTELLI		2024-06-23	
SURFACE FINISH	CHECKED				TITLE
	APPROVED				MAIN HUB ORTHOGRAPHIC
DO NOT SCALE DRAWING					
BREAK ALL SHARP EDGES AND REMOVE BURRS					SIZE A3 · DWG NO. 001 · REV.
FIRST ANGLE PROJECTION	MATERIAL VARIOUS		FINISH SMOOTH		SCALE 1:2 · WEIGHT · SHEET 1 of 2



UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN MILLIMETERS ANGULAR = ± °		NAME	SIGNATURE	DATE	
	DRAWN	FRANCESCO RASTELLI		2024-06-23	
SURFACE FINISH	CHECKED				TITLE
	APPROVED				
DO NOT SCALE DRAWING					
BREAK ALL SHARP EDGES AND REMOVE BURRS					SIZE A3 · DWG NO. · REV.
FIRST ANGLE PROJECTION	MATERIAL		FINISH		SCALE 1:2 · WEIGHT · SHEET 1 of 1

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